Type MS · Explain how skin blood flow and sweating control body temperature

Answer 1 of 13 · Question 7 part b · 4 marks

2020 NOV · Paper 01 (regional R) · find part b in this mark scheme extract

7 (a)	Same diagram with the following modifications: • shunt vessel noticeably more narrow(1) • capillaries noticeably wider(1)		2
(b)	Any four from: • capillaries dilated/get wider/vasodilation(1) • to allow greater blood flow nearer to skin surface(1) • shunt vessel constricts/becomes more narrow(1) • to divert more blood through capillaries(1) • more heat lost through skin's surface/by radiation(1)	Allow blood vessels for capillaries Ignore arteries/veins	4
(c)	• (change in body temperature) detected by the hypothalamus(1) • impulses sent to effector/named effector (1) • temperature change reversed/named reverse process (1)	Allow reverse argument for both marking points Allow signals for impulses Allow temperature returns to normal	2

Total for Question 7 = 8 Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 2 of 13 · Question 7 part a · 2 marks

2020 NOV · Paper 01 (regional R) · find part a in this mark scheme extract

7	(a)	Same diagram with the following modifications: • shunt vessel noticeably more narrow(1) • capillaries noticeably wider(1)		2
	(b)	Any four from: • capillaries dilated/get wider/vasodilation(1) • to allow greater blood flow nearer to skin surface(1) • shunt vessel constricts/becomes more narrow(1) • to divert more blood through capillaries(1) • more heat lost through skin's surface/by radiation(1)	Allow blood vessels for capillaries Ignore arteries/veins	4
	(c)	• (change in body temperature) detected by the hypothalamus(1) • impulses sent to effector/named effector (1) • temperature change reversed/named reverse process (1)	Allow reverse argument for both marking points Allow signals for impulses Allow temperature returns to normal	2

Total for Question 7 = 8 Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 3 of 13 · Question 3 part b · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Question number				
3	(a)	any two from • protection/named example/barrier against pathogens; • sensory reception/named example; • excretion		2
	(b)	• looks red; • vasodilation/more blood in skin <u>capillaries</u>/blood flows closer to skins surface; • allows more heat loss/heat energy released; • to lower body temperature;		4
	(c) (i)	• as temperature increases, rate of sweating increases; • up until 34°C/ between 32 °C - 35.9 °C, little/no change; • increases, rapidly/significantly from 35.9 °C - 38 °C;	Allow any temperature between 34°C and 35.9 °C for 35.9 °C e.g. between 32°C and 34 °C	2
	(ii)	• absorb sweat with cotton wool/absorbent paper; • fixed area of skin/under arm pits; • reference to measuring mass of absorbent material before and after; or • measure time; • for cobalt chloride paper to change colour;		2
	(d)	• enzymes affected/to maintain enzyme activity/ref to body temperature being optimum for enzymes; • too high a temperature denatured/prevent denaturation; • too low, inactive/less active/prevent enzymes becoming inactive;		3

Total 13 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 3 of 13 · Question 3 part b · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 4 of 13 · Question 5 part b · 3 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

5 (a)	• collect urine throughout the cold day; • measure the volume produced; • repeat for a hot day; • compare values;		4
(b)	any three from • vasodilation/arteriole widens; • increased flow of blood through skin; • heat lost through the skin's (surface); • by radiation;	allow artery	3
(c)	• (more) ADH released; • from pituitary gland; • makes kidney tubules/collecting ducts (more) permeable; • (more) water reabsorbed into the body/blood/from filtrate/urine;		4
			Total 11

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 4 of 13 · Question 5 part b · 3 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 5 of 13 · Question 3 part b.ii · 4 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

3 (a)	(i)	250:750; 1:3;	full marks for correct answer with no working	2
	(ii)	17 500 - 5600 = 11900; 11 900 × 100; 5 600 = 210%;	full marks for correct answer with no working 213/212.5 = two marks	3
	(iii)	• other parts of the body/organs; • liver/kidney/lungs/arteries/veins; • needs some blood supplied/blood flows to these organs;		3
(b)	(i)	• large/big/huge increase in total flow; • large/big/huge increase in flow skeletal muscles/skin; • decrease for gut;		3
	(ii)	• heat generated(body) temperature increases during exercise; • from respiration; • carried by blood/more blood to skin; • (heat) lost through skin/by radiation;		4

Total 15 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 5 of 13 · Question 3 part b.ii · 4 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 6 of 13 · Question 8 part b.i · 3 marks

2024 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

8 (a)	any three from • released from pancreas; • when glucose levels rise; • causes/stimulates conversion of glucose to glycogen; • in liver/muscles;		(3)
(b) (i)	• vasodilation; • more blood flows at/near the surface of the skin; • heat (from blood) radiated (away from body)/increased heat loss; • hairs lie flat; • less air trapped; • (more) heat radiated;		(3)
(ii)	vasoconstriction/blood vessels in the skin narrowed shivering/increased metabolic rate/hairs stand on end;		(1)
(iii)	• hypothalamus detects rise/fall/change in body temperature; • (named) mechanism reverses the change/brings back to set point/normal;	e.g. vasodilation, increase in sweating	(2)
(c)	• enzymes lose kinetic energy; • enzyme activity decreases/works slowly; • less enzyme substrate complexes formed/ less collisions between enzyme and substrate/binding with substrate; • rate of chemical reactions/metabolism slow down; • reference to symptom of hypothermia e.g. death/weak pulse/slower heart/breathing rate;		(5)

Total for Question 8 = 14 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 7 of 13 · Question 8 part b.ii · 1 mark

2024 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

8	(a)	any three from • released from pancreas; • when glucose levels rise; • causes/stimulates conversion of glucose to glycogen; • in liver/muscles;		(3)
	(b) (i)	• vasodilation; • more blood flows at/near the surface of the skin; • heat (from blood) radiated (away from body)/increased heat loss; • hairs lie flat; • less air trapped; • (more) heat radiated;		(3)
	(ii)	vasoconstriction/blood vessels in the skin narrowed shivering/increased metabolic rate/hairs stand on end;		(1)
	(iii)	• hypothalamus detects rise/fall/change in body temperature; • (named) mechanism reverses the change/brings back to set point/normal;	e.g. vasodilation, increase in sweating	(2)
	(c)	• enzymes lose kinetic energy; • enzyme activity decreases/works slowly; • less enzyme substrate complexes formed/ less collisions between enzyme and substrate/binding with substrate; • rate of chemical reactions/metabolism slow down; • reference to symptom of hypothermia e.g. death/weak pulse/slower heart/breathing rate;		(5)

Total for Question 8 = 14 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 8 of 13 · Question 3 part b · 6 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

3 (a)	• control of/maintenance of • constant internal environment; • despite changes in external environment; • ref negative feedback;	ignore specific examples	4
(b)	box 1 hypothalamus; detects rise/increase in blood temperature; box 2 arterioles dilate/vasodilation; box 3 sweat evaporates; box 4 (more) heat loss (from skin); by radiation;	ignore thermoregulatory centre	6
(c)	• enzymes; • denatured; • fewer/no enzyme substrate complexes formed/substrate can't bind to active site/reduced enzyme activity/reaction; • reduced metabolic activity/rate of reactions in body reduced;		4
			14 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 8 of 13 · Question 3 part b · 6 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 9 of 13 · Question 5 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

5(a)	• correct labels on correct axes; • correct/appropriate scale; • correct plotting; • suitable lines; • lines labelled;	Allow line of best fit	**(5)**

Question Number	Answer	Additional guidance	Mark
5(b)	12.5 - 2.5; 10;	ECF from 5a	**(2)**

Question Number	Answer	Additional guidance	Mark
5(c)	Any three from: • exercise requires more energy; • from respiration; • more heat released raising body temperature; • more sweat released to cool body;	Reject energy produced/made Ignore anaerobic respiration	**(3 max)**

Total 10 Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 9 of 13 · Question 5 part c · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 10 of 13 · Question 3 part c · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number			
3 (a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
(e)	obesity / high sugar intake / high BMI;		1
(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 10 of 13 · Question 3 part c · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 11 of 13 · Question 3 part f · 5 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part f in this mark scheme extract

number			
3 (a)	C hypothalamus; A does not control sweating B does not control sweating D does not control sweating		1
(b)	B motor; A transmits impulse from sensory to motor C alternative name for A with same function D transmits impulse to brain		1
(c)	• (sweat) evaporates from (the surface of) the skin; • using heat from body/blood; • cools body/reduces body temperature;		3
(d)	• runs in families; • must be faulty/mutated gene/chromosome/allele; • which transfers/passes on information to offspring;		3
(e)	obesity / high sugar intake / high BMI;		1
(f)	• exercise/activity of individual varies; • greater rate of aerobic respiration; • heat given off as by-product/increased body temperature; • external temperature varies; • hotter more sweating/cooler less sweating; • mass of individual varies; • volume of water in the body/fluid intake varies;	Allow reverse argument for mps 2 and 3	5 Total 14

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 11 of 13 · Question 3 part f · 5 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part f in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 12 of 13 · Question 5 part b.ii · 5 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number					
5	(a)	(i)	any two from • to allow the heart rate to return to normal; • allow respiration/breathing rate/body temperature to return to normal; • to allow sweat to evaporate;		2
		(ii)	any three from • same time for exercise; • same type/intensity of exercise; • wear same clothing for each test; • measure sweat from same area;		3
	(b)	(i)	• axes labelled with units with independent variable on x-axis; • suitable scale; • both sets of data plotted correctly;; • suitable lines; • lines labelled;		6
		(ii)	any five from • exercise increases muscle activity; • more energy/ATP required; • increased respiration; • heat produced as by product; • lost from body; • by increased sweating; • increase in room temperature increases rate of sweating;		5

Total 16 marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 12 of 13 · Question 5 part b.ii · 5 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how skin blood flow and sweating control body temperature

Answer 13 of 13 · Question 1 part b.ii · 2 marks

2024 JUN · Paper 01 · find part b.ii in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks